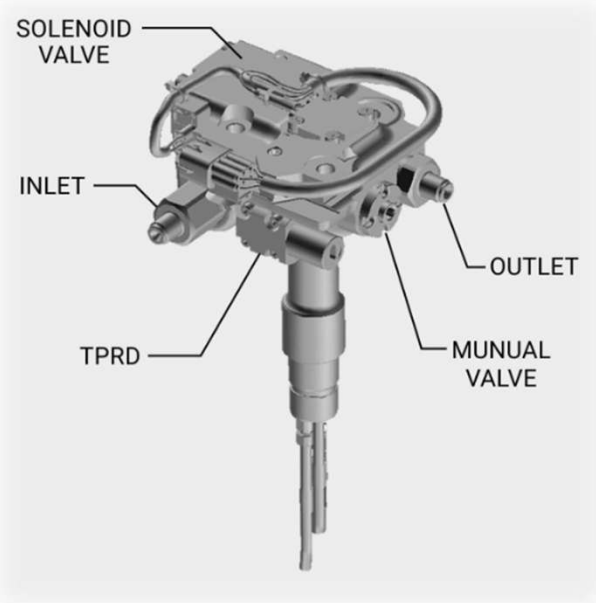
SELECTIVE ASSEMBLY OF RECOVERED HYDROGEN

Valve Components



Haed Tavakkoli-moghaddam - October 25

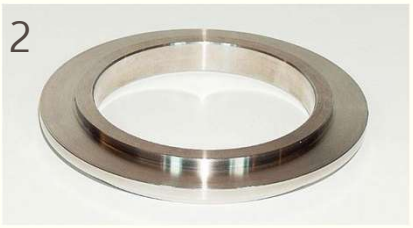
Title and Content Layout with List

- Problem Description
- Engineering Model
- Mathematical Model
- Experiments
- Result
- Conclusion

Problem Description

- A manufacturer of high-pressure hydrogen valves with two sources of components:
    1. New components
    2. Recovered components
- Recovered components cheaper, but different dimensions from new components. So, they cannot be assembled interchangeably.
- Research question: Is it possible to select combinations of components that satisfies dimensional quality while minimizing manufacturing cost?
- Three components is considered (JTEKT hydrogen-valve product)[1]:
    1. Valve body
    2. Valve-seat insert
    3. Plunger

1 

2 

3 

[1] https://www.jtekt.co.jp/e/products/hydrogen/valve01.html

Engineering Model

- First interface: valve body and seat:

$$I_{ij} = D_j^s - D_i^p$$
$$0.006 \leq I_{ij} \leq 0.020$$

- Second interface: seat and plunger:

$$O_{ijkr} = L_k + H_j - r - P_i$$
$$0.025 \leq O_{ijkr} \leq 0.083 \text{ mm}$$

Where L_k, H_j, r, and P_i respectively are plunger closing reach, seat height, seat-height reduction, and body pocket depth

- For recovered components two alternative for restoration is considered:

$$r \in \{0.015,\ 0.030\} \text{ mm}$$

- Synthesis data uncertainty is considered because the actual diameter may differ from measured numbers so:

$$U_I = u_{s,out} + u_{b,bore}$$
$$U_O = u_p + u_{s,height} + u_{b,depth} + u_{rework}$$

And the assembly would be accepted when:

$$I_{\min} \leq I - U_I \ \ \& \ I + U_I \leq I_{\max}$$
$$O_{\min} \leq O - U_O \ \ \& \ O + U_O \leq O_{\max}$$

Optimization Model

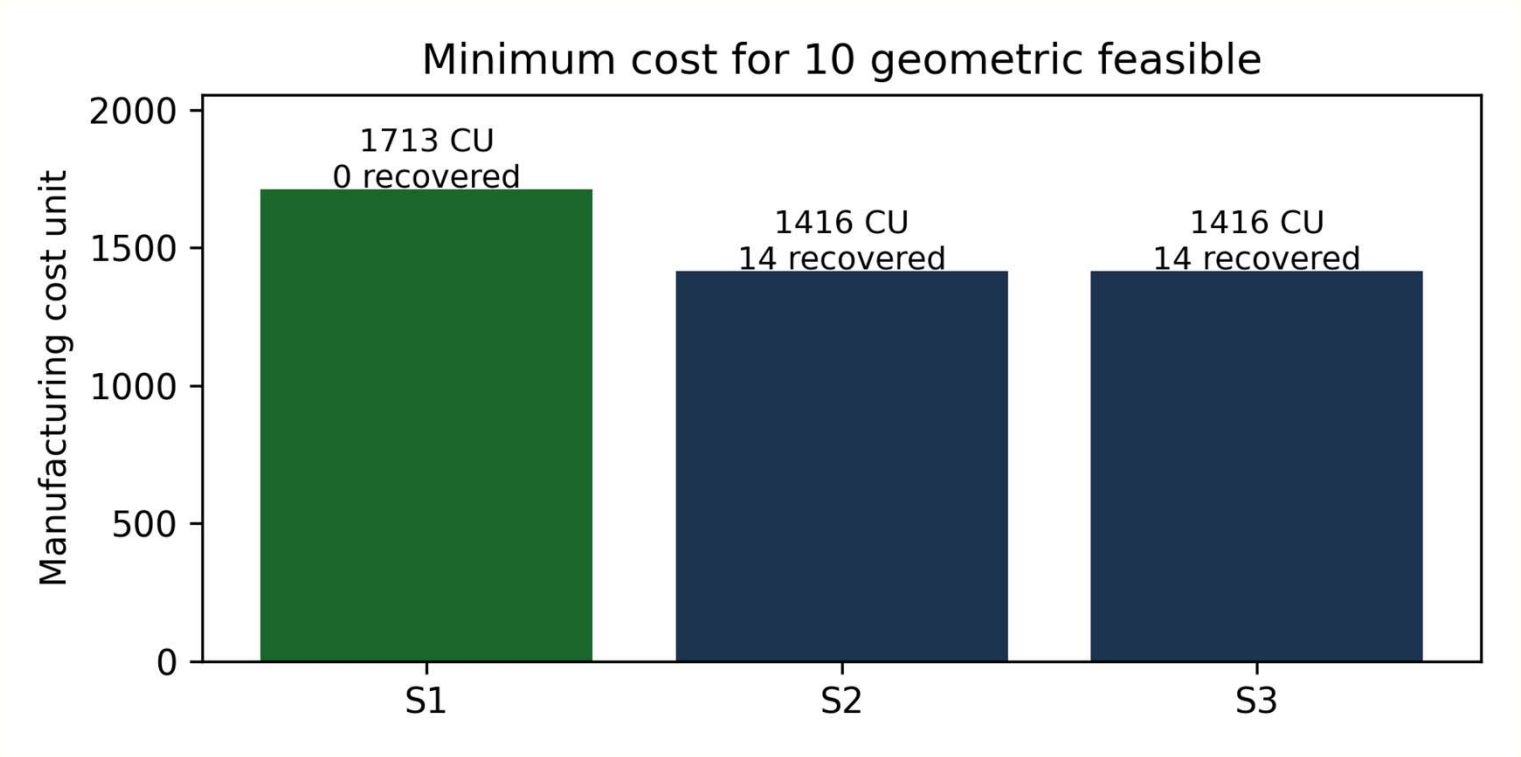
- Binary variable: $x_{ijkr} = 1$ if assembly (i, j, k) is selected with finishing option r
- Objective function:

$$\min Z = \sum_{i,j,k,r} c_{ijkr}\, x_{ijkr}$$

- Constraints:
  - Production demand considered
  - Each component used once
  - Robust interference and overtravel
  - Reworked seats for capacity

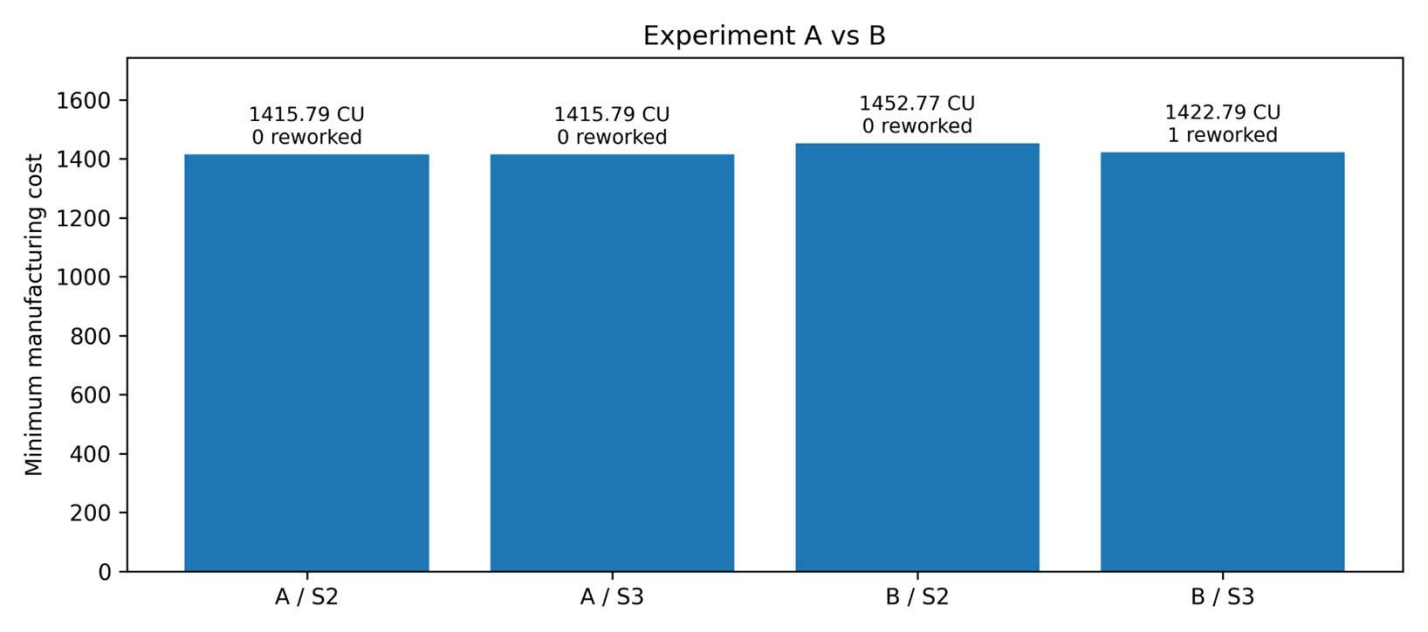
Experiment 1: rework not necessary

- Observation: recovered components reduce manufacturing cost by 17% compare to new component



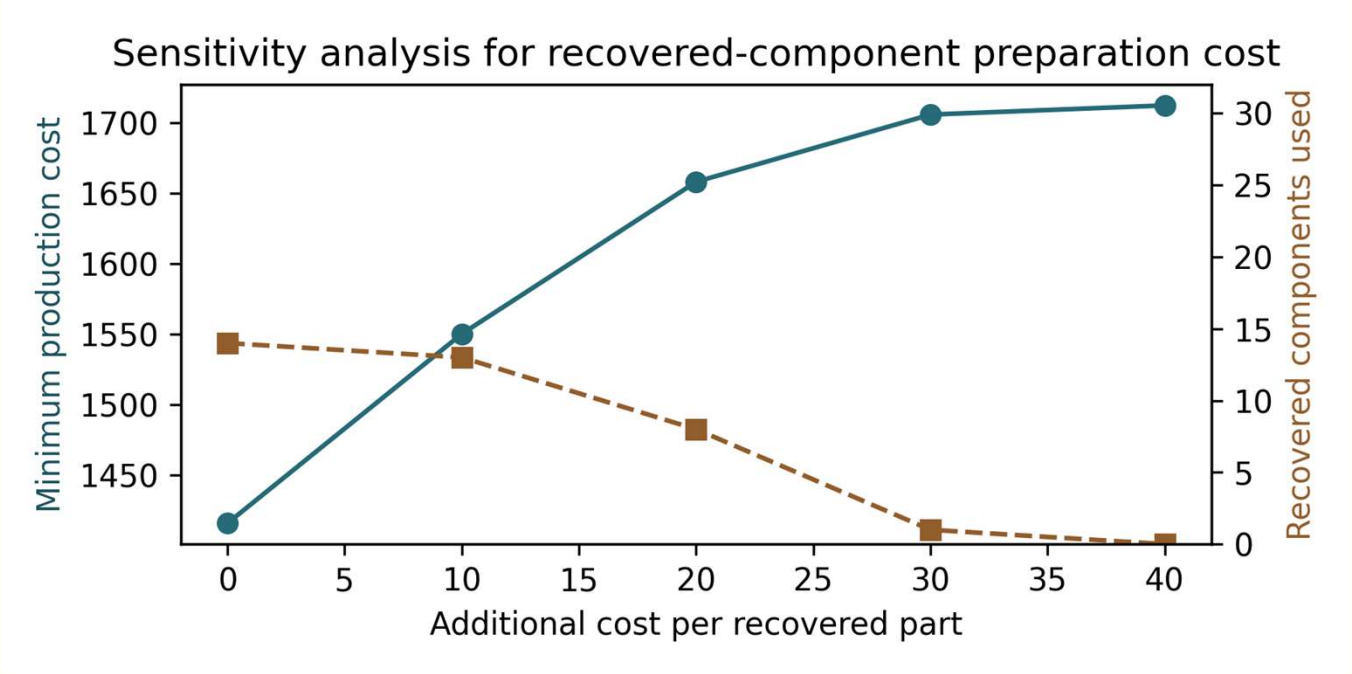
Experiment 2: rework necessary

- Experiment 2: testing if seat height increment will change the result or not.
- Observation: seat rework is feasible for assembly and cheaper. The economic value depends on the condition of components and geometric compatibilities with the consideration of costs for recovery and finishing.



Sensitivity Analysis: recovery-cost

- If the cost of preparing recovered components increases, the model replace recovered components with new ones. So, we can have a trade-off between cost minimization and reuse maximization. The problem can be considered as a multi objective problem.



Conclusion and Future Research

- Selective assembly reduced cost by 17% in reference case.
- Adaptive seat rework was not necessary in first experiment, but because of geometric values changes in second experiment was beneficial.
- The economic advantage of recovered components depends on dimensional compatibility and recovery cost.
- For future research an adaptive decision-making can be used with predication models.
- Another future research direction can be considering quality control in the model with considering leakage and sealing performance.